

Section 2 – Shear and Moment Diagrams

Provide information for constructing the **shear diagram**. The x origin is on the left at A and the support reactions are $A_y = 6$ kip acting up, $C_y = 9$ kip acting up and $M_C = 60$ kip-ft acting clockwise. A is a roller, B is a pin and C is a fixed support.

4. (3 pt) What happens to the shear diagram at $x = 12$ ft?

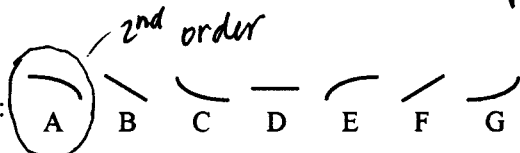
- A. No change
B. Decrease 9 kip
C. Increase 9 kip
D. Decrease 3 kip
E. Increase 3 kip
F. Decrease 1.5 kip

Determine the numerical value (magnitude only) of the change in shear for section $0 < x < 12$ ft as well as the behavior (shape) of the shear for the same region.

5. (3 pt) Change = -9 kip (magnitude only)

- A. 9
B. 1.5
C. 6
D. 12
E. 18
F. 0

6. (3 pt) Behavior:



7. (3 pt) What happens to the shear diagram at $x = 16$ ft?

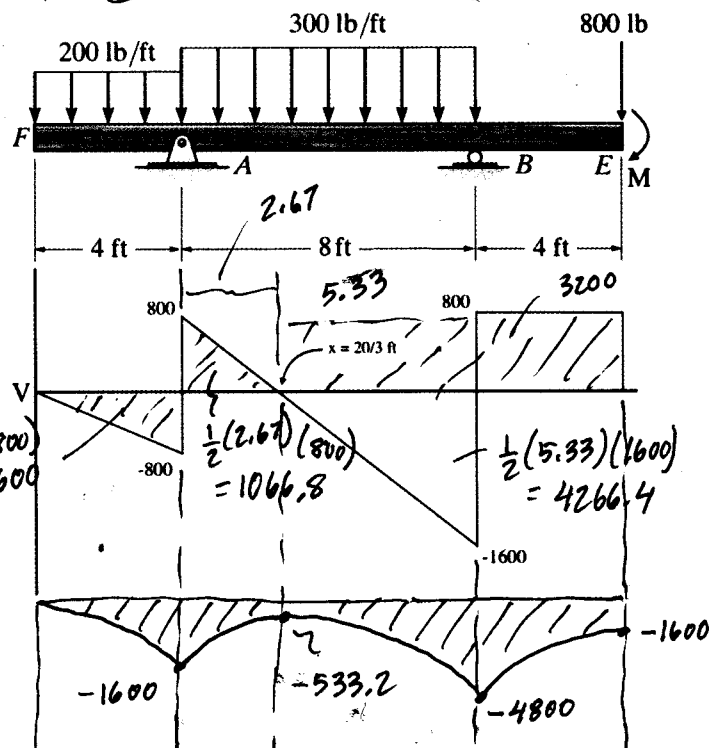
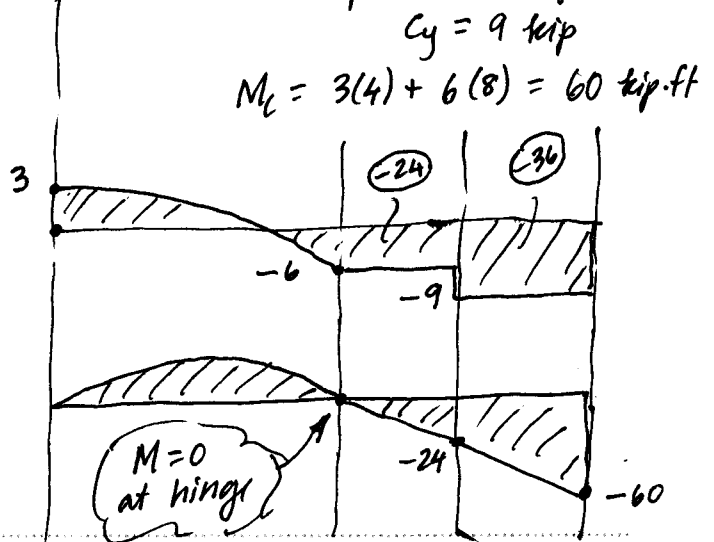
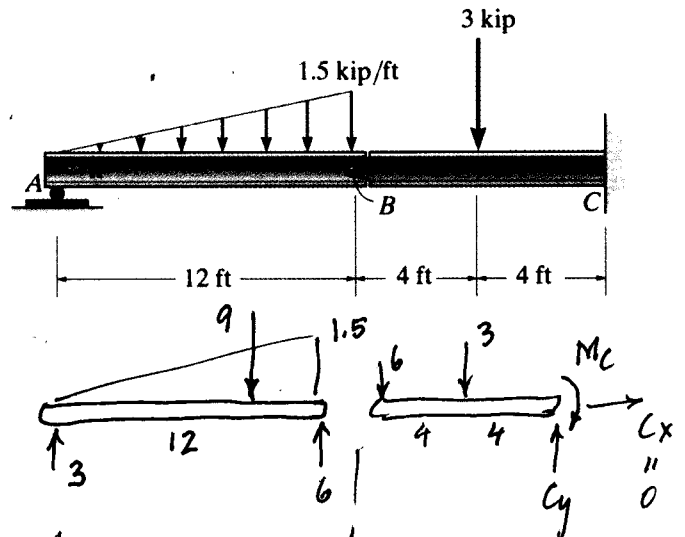
- A. Decrease 3 kip
B. Increase 3 kip
C. Decrease 9 kip
D. Increase 9 kip
E. Decrease 6 kip
F. Increase 6 kip

Provide information for constructing the **moment diagram**. The x origin is on the left at A and the support reactions are $A_y = 1600$ lb acting up, $B_y = 2400$ lb acting up. A is a pin and B is a roller. The couple $M = 1600$ lb-ft CW at E.

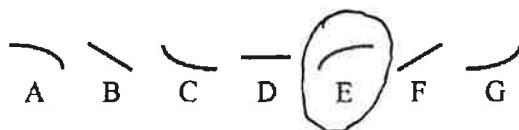
Determine the numerical value (magnitude only) of the change in moment for section $4 < x < 6.667$ ft as well as the behavior (shape) for the same region.

8. (3 pt) Change = _____ lb-ft (magnitude only)

- A. 1067
B. 800
C. 1600
D. 2400
E. 4267
F. 3200



9. (3 pt) Behavior:



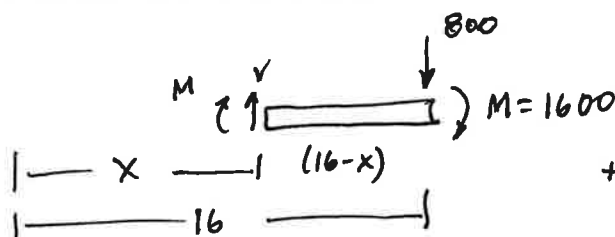
10. (3 pt) What happens to the moment diagram at $x = 16$ ft?

- A. Increase 1600 lb·ft
B. Decrease 400 lb·ft
C. Increase 400 lb·ft
D. Decrease 1600 lb·ft
E. Increase 800 lb·ft
F. No change



11. (4 pt) Determine the order of the equation for the internal bending moment for the region $12 < x < 16$ ft on the beam shown.

- A. third order, cubic
B. constant
C. second order, parabolic
D. zero
E. first order, linear



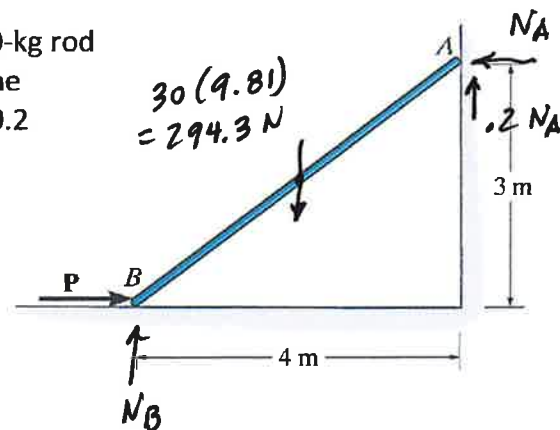
$$+\circlearrowleft \sum M = 0; \quad M + 800(16-x) + 1600 = 0$$

$$M = -800(16-x) - 1600$$

Section 3 - Friction

12. (5 pt) Determine the minimum force P to prevent the uniform 30-kg rod AB from sliding down. The contact surface at B is smooth, whereas the coefficient of static friction between the rod and the wall at A is $\mu_s = 0.2$

- A. 143 N
B. 155 N
C. 138 N
D. 129 N
E. 117 N



$$+\circlearrowleft \sum M_B = 0; \quad -294.3(2) + N_A(3) + 0.2N_A(4) = 0$$

$$N_A = 154.9 \text{ N}$$

$$P = N_A$$